

384*

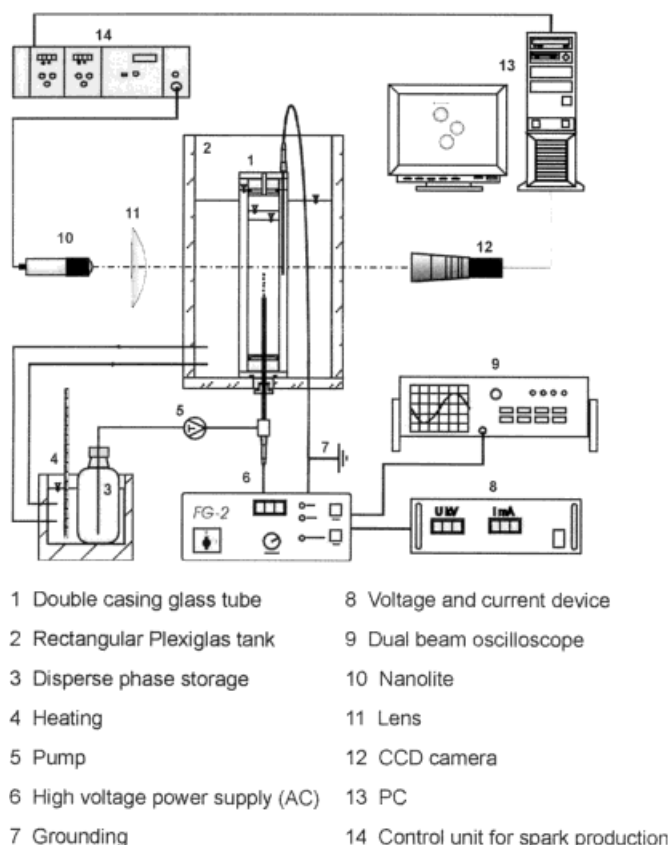
Efficient Generation of Mono-dispersed Particles Using High Frequent Electric Fields

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Considering the droplet formation process at a capillary, it is possible to control the droplet size by applying an external electric field to the capillary. If the forming droplet has reached its charge limit, which depends on the applied voltage, the permittivity of the surrounding medium, the interfacial tension of the two liquids and the radius of the droplet, it is forced to detach earlier or to disintegrate into a large number of daughter droplets. Exploiting this, a set of experiments were carried out, where the formation of mono-sized droplets down to $90\text{ }\mu\text{m}$ was possible in different liquid-liquid systems. The droplet formation process was studied under field conditions, where the high frequent electric field (AC, 30 and 45 kHz) was varied in the range of 0 to 20 kV. As a result, between all the varied parameters the applied voltage had the strongest influence on droplet size.

Figure.
Experimental set-up.



385*

Investigation of the Shape and Dynamic of Fast Microparticles with a Single-Shot-Micro-CCD

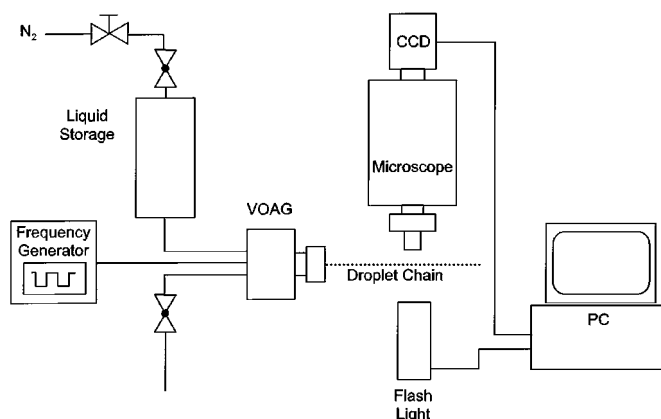
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Several aspects of the formation, dynamic and interaction of airborne microparticles can be well investigated observing monodisperse and equidistant droplet chains of different liquids or solutions. The variability of droplet chains is limited by minimum initial velocity, feasible frequency ranges and restricted coherence length. Investigations are further constricted by the ability of the observation methods used.

The actual experimental set-up consists of a Berglund-Liu generator fixed on a software driven scanning table, a light microscope with a long working distance objective combined with a CCD-camera and a short exposure sparkflash light. With the present investigations a correlation of frequency dependent jet disintegration and the electrical charge of the droplets could be evidenced.

Figure.
Experimental set-up.



386*

Split of a Pneumatic Conveying Line to Multiple Parallel Lines – Prediction of Mass Flow Distribution and Control

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Distribution of pulverized material, e.g. pulverized coal, from one conveying line evenly to several lines is a problem of growing interest. In such a system the material will be distributed from the main pipe to the subsequent lines according to the individual conveying resistances. In the present work a calculation procedure has been developed, that uses individual models for different types of pipe sections. These submodels have been verified by experimental work. The model allows the determination of the distribution for a given system of lines and a given total injection rate. In order to provide a system for dynamic adjustment, it is necessary to vary the conveying resistance of the lines. Therefore, various mechanical valves as well as addition of secondary air have been tested. Appropriate submodels were included in the model approach.

387

Flame Generated γ -Fe₂O₃ Nanoparticles: Formation Kinetics and Particle Characteristics

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The understanding of process parameter and their influence on the properties of the respective product is essential in nanoparticle formation. Thus, the application of appropriate measuring techniques are inevitable for the design of powders with specific properties. In this work, the formation and growth of γ -Fe₂O₃ nanoparticles has been followed in-situ by particle mass spectrometry. Particle formation in a low-pressure flame reactor utilizing a hydro-oxygen flame doped with iron-pentacarbonyl (Fe(CO)₅) as precursor and the influence of process parameters as temperature, pressure, and precursor concentration on particle size and size distribution parameter was studied. Ex-situ measurements of the specific surface area, the crystallite structure, and the magnetic properties yield insight into the characteristic of nano-sized γ -Fe₂O₃. It has been found that the residence time in the flame is the most mass determining parameter whereas the influence of the flame condition is very weak. Furthermore, the experimental findings were validated applying a comprehensive computer model including homogeneous chemical reactions of the burner stabilised flames as well as the particle dynamics due to Brownian motion. The results of both, the measurements and the computation are in good agreement.

388

Flame Synthesis of Nanoparticles

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An overview of recent advances in synthesis of nanoparticles by flame aerosol processes is given. In flame processes with gaseous precursors emphasis is placed on reactant mixing and composition, additives, and external electric fields for control of product characteristics. The formation and growth of nanoparticles can be monitored by thermophoretic sampling, while the corresponding temperature history can be obtained by non-intrusive techniques. Furthermore, applications of flame-made nanoparticles for reinforcement or catalysis are discussed at production rates up to 700 g/h. In flame processes with liquid precursors by the so-called flame spray pyrolysis (FSP), emphasis is placed on reactant and fuel composition for control of product powder characteristics. The FSP processes are quite attractive as they can employ a wide array of precursors, so a broad spectrum of new nanosized powders can be made.

389

Simultaneous Assessment of Particle Size Distributions and Gas Temperatures from Time-Resolved Laser-Induced Incandescence

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Point measurements of time-resolved LII signals have been performed in sooting premixed low-pressure flames. Soot particle size distribution and gas temperature in these flames are known from independent measurements. This data is used to validate parameters in an improved LII model. Special emphasis is taken on the accurate modelling of mass and heat transfer rates. Using this model particle size distributions and gas temperatures can be estimated from time-resolved LII signals using nonlinear regression. Standard numerical methods are applied. An experimental setup is presented, which allows measuring one-dimensional maps of particle size distribution and gas temperature. The technique is based on the one-dimensional and time-resolved detection of LII-signals using a Streak camera.